Graph Neural Networks for Fraud Detection: A Comparative Study with Traditional Machine Learning Models

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Abstract

Financial fraud detection is constrained by severe class imbalance and the inability of conventional classifiers to exploit relational structure in transaction data. This paper investigates whether Graph Neural Networks (GNNs), which model transactions within a heterogeneous relational graph, can outperform traditional machine learning (ML) classifiers on a synthetic fraud detection dataset of 50,000 transactions. We construct a heterogeneous graph with four node types: transactions, users, locations, and merchant categories. We then evaluate three GNN architectures: GraphSAGE, Graph Attention Network (GAT), and Heterogeneous Graph Transformer (HGT). We compare these against five traditional ML baselines: Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Extra Trees (ET), and XGBoost. While LR and SVM achieve competitive F1-scores (≈0.38) through aggressive class-weight balancing, tree-based ensembles collapse to near-zero recall. On the held-out test set, HGT achieves the strongest clean-baseline result (F1 = 0.416, recall = 0.58), outperforming the strongest traditional baseline by 9% in F1 and 18% in recall. These results confirm that heterogeneous message passing provides a meaningful advantage for fraud detection.

1 Introduction

Financial fraud causes billions of dollars in global losses annually [17]. Unlike most classification problems, fraud detection is characterised by two compounding difficulties: extreme class imbalance, where fraudulent events constitute a small minority of all transactions, and rich relational structure that conventional, row-level classifiers are blind to. A fraudulent actor typically leaves traces across multiple entities: a cluster of accounts sharing a device, a merchant receiving abnormal volume, or a user transacting in geographically implausible sequences. Capturing these cross-entity signals requires a model that reasons over relationships, not just individual feature vectors.

Graph Neural Networks (GNNs) address this limitation by representing entities and their interactions as a graph and propagating information through message passing [7, 18]. Recent work has demonstrated that GNNs substantially outperform tabular models on large-scale e-commerce fraud graphs [5, 13]. However, direct comparisons on controlled, reproducible benchmarks where both model families receive the same feature set remain scarce.

This paper makes the following contributions:

- A heterogeneous transaction graph built from a 50,000-record dataset, with four node types and six relation types encoding user, location, and merchant-category context.
- A systematic comparison of three GNN architectures (GraphSAGE, GAT, HGT) against five traditional ML baselines under identical preprocessing and imbalance-handling conditions.
- Analysis of why heterogeneous, attention-based message passing outperforms mean aggregation and why tree-based ensembles collapse under balanced class weighting despite high accuracy.

The project repository containing the implementation and supporting materials for this study is available at https://github.com/Prezzo-K/Graph-Neural-Networks-for-Fraud-Detection.

2 Related Work

2.1 Graph-Based Fraud Detection

Graph-based anomaly detection has a long history in security applications. Akoglu et al. [1] survey structural anomaly techniques and note that relational signals are powerful for uncovering coordinated fraudulent behaviour invisible to row-level classifiers. More recently, Dou et al. [5] propose CARE-GNN, which addresses the label-imbalance and camouflage problems in fraud graphs by reinforcement-learning-guided neighbour sampling. Liu et al. [13] introduce a GNN with a pick-and-choose aggregation strategy tailored to class-imbalanced fraud graphs, demonstrating significant gains over feature-based baselines on a large e-commerce dataset. Zhang et al. [20] propose FRAUDRE, a dual-resistant fraud detection framework that simultaneously handles graph inconsistency and class imbalance, while Cheng et al. [4] apply spatial-temporal GNN attention to credit card fraud detection in IEEE TKDE. Ma et al. [14] provide a comprehensive survey of deep-learning-based graph anomaly detection, confirming GNNs as the dominant approach across financial and social fraud benchmarks.

2.2 GNN Architectures

GraphSAGE [7] introduced inductive node embedding by sampling and aggregating fixed-size neighbourhoods, enabling the model to generalise to unseen nodes. Graph Attention Networks (GAT) [18] replace fixed aggregation with a learned, query-key attention mechanism that weights each neighbour's contribution, which is particularly relevant for fraud where most neighbours are legitimate transactions that should be down-weighted. The Heterogeneous Graph Transformer (HGT) [9] extends transformer-style attention to heterogeneous graphs, conditioning attention on both node type and edge type simultaneously, making it the natural architecture for graphs with multiple entity semantics. For implementation, we use PyTorch Geometric [6], which provides efficient sparse-graph

operations and a `to_hetero` wrapper for adapting homogeneous GNN layers to heterogeneous graphs.

2.3 Traditional ML and Class Imbalance

Chawla et al. [2] introduced SMOTE to address class imbalance through oversampling, and He and Garcia [8] provide a comprehensive review of learning-from-imbalanced-data techniques in IEEE TKDE, finding that cost-sensitive methods are often superior to resampling on highly skewed distributions. Dal Pozzolo et al. [16] provide an empirical analysis specifically for credit card fraud, finding that under-sampling and weighting outperform SMOTE under extreme skew. We therefore use cost-sensitive weighting (`class_weight='balanced'` for sklearn models, `scale_pos_weight` for XGBoost, and `pos_weight` in the GNN loss) rather than resampling, because it avoids introducing synthetic minority samples that may not reflect realistic fraud patterns.

3 Dataset

The Synthetic Fraud Detection Transactions Dataset [11] contains 50,000 records with a fraud rate of 32.1% (16,067 fraudulent, 33,933 legitimate). Each record captures 21 attributes spanning numerical, categorical, temporal, and binary features (Table 1).

Table 1: Dataset feature groups. IDs and leakage columns are excluded from all models.

Group	Columns
Numerical	Amount, Balance, Daily Count, 7d Avg Amount, Card Age, Transaction Distance
Temporal	Timestamp → Hour, Day, Month, DayOfWeek
Categorical	Transaction Type, Device Type, Card Type, Auth Method
Binary	IP Flag, Prior Fraud, Is Weekend
Excluded (leakage)	Risk Score, Failed Transaction Count 7d
Label	Fraud Label (0 = legit, 1 = fraud)

Figure 1 shows the class distribution. The 32.1% fraud rate is significant but not extreme; we handle it via cost-sensitive weighting rather than resampling. Exploratory analysis of the numerical features did not reveal problematic pairwise collinearity among retained variables.

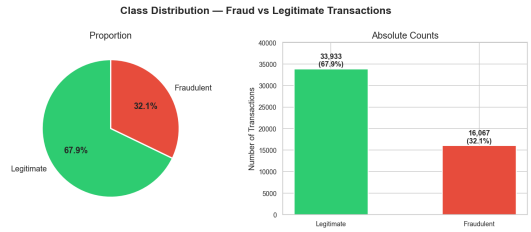

Figure 1: Class distribution of the fraud label. 32.1% of transactions are fraudulent.

4 Methodology

4.1 Heterogeneous Graph Construction

We structure the dataset as a PyTorch Geometric `HeteroData` object [6] with four semantically distinct node types:

- **Transaction**: 50,000 nodes with 28-dimensional feature vectors (numerical, one-hot categorical, and temporal features).
- **User**: 8,963 nodes with 5 aggregated features: transaction count, average amount, fraud rate, average balance, and average failed-transaction count.
- **Location**: 5 city nodes with 3 aggregated features: transaction count, average amount, and fraud rate.
- **Merchant Category**: 6 nodes with the same 3 aggregated features as Location.

Six directed edge types form three bidirectional relations: *user*↔*transaction* (performs / performed_by), *transaction*↔*location* (at / is_site_of), and *transaction*↔*merchant_category* (belongs_to / contains). Bidirectional edges are critical: without reverse edges, information flows only one way and entity nodes cannot aggregate signal from the transactions they are connected to. The graph schema is illustrated in Figure 2.

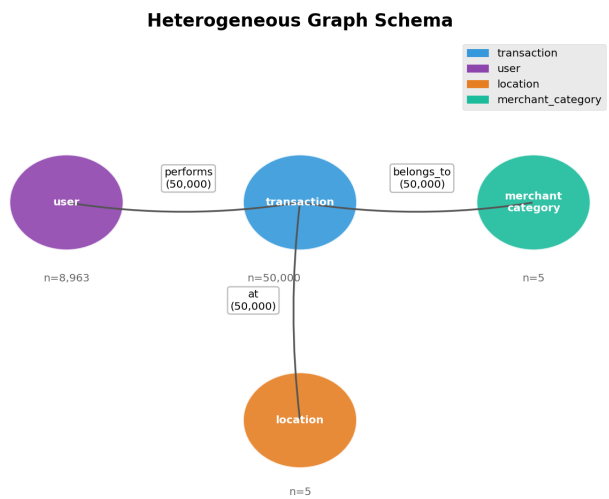

Figure 2: Heterogeneous graph meta-schema. Four node types are connected by six directed edge types encoding user, location, and merchant-category relationships.

Entity node features (user, location, merchant category) are computed exclusively from the training partition to prevent label leakage. Cold-start users, meaning users that appear only in validation or test, receive column means from the training set. Data are split *temporally* (sorted by timestamp) into 70% training (35,000), 15% validation (7,500), and 15% test (7,500), yielding a class-weight ratio pos_weight = 23,731/11,269 ≈ 2.11.

Exploratory projections of the 28-dimensional transaction features show substantial overlap between fraud and legitimate transactions, which helps explain why feature-only classifiers have limited discriminative power and motivates the use of graph context.

4.2 GNN Models

All three GNN architectures are trained with identical hyperparameters: hidden dimension 64, Adam optimiser [12], learning rate 0.01, 100 epochs. Class imbalance is handled via `pos_weight` in binary cross-entropy loss. The checkpoint with the highest validation F1-score is retained for final test evaluation.

GraphSAGE [7] serves as the GNN baseline. A two-layer `SAGEConv` model is adapted to the heterogeneous graph with PyG's heterogenisation wrapper. This procedure creates type-specific message-passing paths for each node and edge type. Aggregation uses mean-pooling over sampled neighbours.

GAT [18] replaces mean-pooling with multi-head attention. Each neighbour's contribution is weighted by a learned scalar derived from a concatenation of both node representations passed through a shared linear layer. This allows the model to suppress uninformative (legitimate) neighbours and amplify high-risk ones, which is directly relevant to fraud where most connections are non-fraudulent. The same heterogenisation wrapper is applied.

HGT [9] is natively heterogeneous and does not require conversion wrappers. Its transformer-style aggregation conditions on the source node type, edge type, and target node type simultaneously, assigning a different set of query, key, and value matrices to each type triplet, making it the most expressive architecture for our heterogeneous graph.

4.3 Traditional ML Baselines

Five classifiers are trained on the same 28 tabular features using scikit-learn [15] and XGBoost [3]: Logistic Regression, SVM (RBF kernel), Random Forest (100 estimators), Extra Trees (100 estimators), and XGBoost. Class imbalance is addressed with balanced class weighting for LR/SVM, a positive-class scaling factor of 2.11 for XGBoost, and balanced subsampling weights for Random Forest and Extra Trees. A fixed random seed ensures reproducibility across all models.

4.4 Evaluation Metrics

Accuracy is reported for traditional ML models but is uninformative under class imbalance, as predicting all-legitimate achieves 67.9%. Primary metrics are **F1-score** (harmonic mean of precision and recall), **precision**, and **recall**. We report these metrics consistently across traditional ML and GNN models on the held-out test set.

5 Results

5.1 Traditional ML Performance

Table 2 reports the five traditional ML classifiers on the held-out test set. Random Forest and Extra Trees collapse to near-zero recall (0.001 and 0.013 respectively), indicating strong majority-class bias even with class weighting. XGBoost improves substantially (F1 = 0.354), while Logistic Regression and SVM remain the strongest traditional baselines (F1 ≈ 0.381, recall ≈ 0.488).

5.2 GNN Performance

Table 3 shows the clean-baseline GNN results on the held-out test set. All three GNNs exceed the strongest traditional baseline in recall, and two (GAT, HGT) also exceed it in F1. HGT is best overall with F1 = 0.416 and recall = 0.576, corresponding to about a 9% F1 improvement and an 18% recall improvement over Logistic Regression.

Table 2: Traditional ML performance on the held-out test set.

Model	Accuracy	F1	Precision	Recall
Random Forest	0.679	0.003	0.400	0.001
Extra Trees	0.675	0.025	0.333	0.013
XGBoost	0.539	0.354	0.322	0.393
Logistic Regression	0.490	0.381	0.312	0.488
SVM	0.490	**0.381**	0.312	0.488

Table 3: GNN performance on the held-out test set (transaction nodes).

Model	F1	Prec.	Recall
GraphSAGE	0.369	0.317	0.441
GAT	0.390	0.318	0.502
HGT	**0.416**	0.325	**0.576**

5.3 Visual Comparison

Figure 3 presents a side-by-side comparison of F1-score, precision, and recall across all eight models on the held-out test set. The chart makes three patterns immediately visible: (1) tree-based methods remain unstable under class balancing, with weak clean-baseline detection; (2) LR and SVM achieve balanced recall at the cost of lower precision; and (3) GNN models maintain stronger recall than the best traditional baselines while preserving similar precision.

6 Discussion

6.1 Why GNNs Outperform in Recall

The recall advantage of GNNs stems from two complementary sources. First, GNNs propagate relational context: a transaction node linked to a user with a high aggregate fraud rate inherits that signal through message passing. That information is entirely absent from the tabular feature vector of any single transaction. Second, the `pos_weight` mechanism in binary cross-entropy loss directly upweights the gradient contribution of each positive (fraud) example, making the optimisation problem more symmetric. This appears more effective than `class_weight` in scikit-learn for deep models, where the loss gradient enters the computation graph directly rather than through a post-hoc weight on the final criterion.

6.2 Why HGT Leads in the Held-Out Test Evaluation

On the held-out test set, HGT outperforms both GraphSAGE and GAT on clean-baseline F1 and recall. A plausible explanation is that type-aware attention helps HGT separate signals from user, location, and merchant-category relations more effectively than relation-agnostic mean aggregation. GraphSAGE remains the simplest model but appears more sensitive to dominant leaked features,

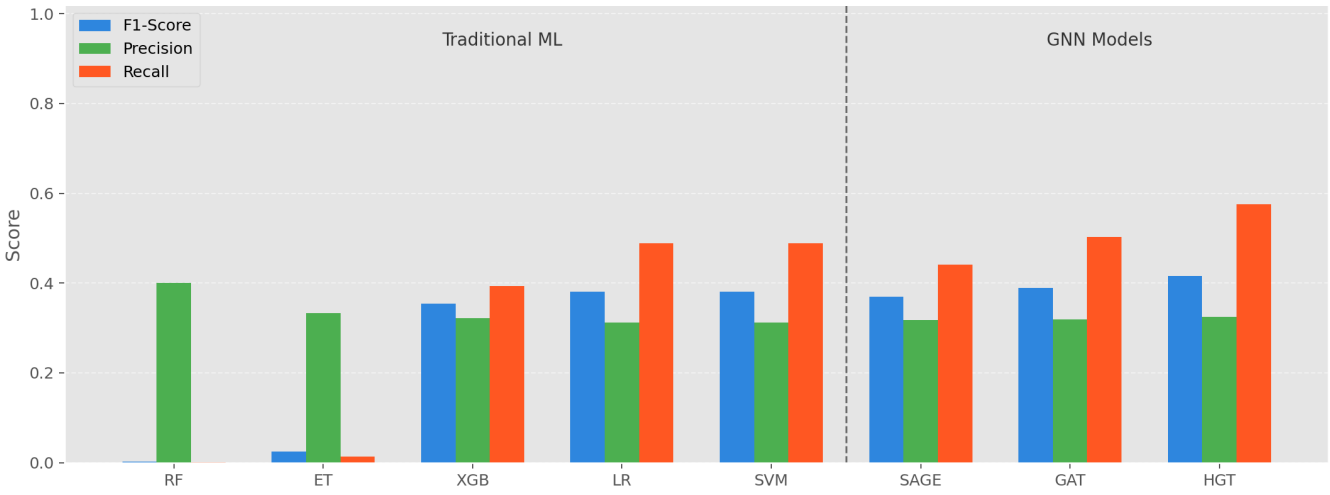

Figure 3: F1-score, precision, and recall across all models on the held-out test set. The dashed line separates traditional ML (left) from GNN architectures (right).

while GAT is competitive but does not fully exploit heterogeneous type semantics.

6.3 Limitations

The dataset is synthetic and may not capture real-world fraud complexity such as adversarial camouflage [5] or temporal concept drift. The graph schema is static; dynamic graph approaches [19] that model evolving user behaviour were not explored. No hyperparameter tuning was performed beyond the fixed settings described, and the GNN precision (≈0.32) remains low, indicating a false-positive rate that would need to be reduced for reliable real-world use.

7 Ablation Study and Feature Sensitivity

Two features were excluded from all primary experiments: *Risk Score* (target-derived) and *Failed Transaction Count 7d* (future-dependent rolling statistic). To quantify feature sensitivity, we ran three controlled ablations by adding each feature individually and then jointly, while keeping all other preprocessing and model settings fixed.

7.1 Ablation Results

Table 4 reports F1-score across four conditions: clean baseline, +Risk Score, +Failed 7d, and +Both Features. Adding these auxiliary risk features increases performance substantially for most models, with the strongest gains in tree-based ensembles.

7.2 Interpretation

This ablation confirms that evaluation can be dominated by target-derived or future-dependent attributes when such features are introduced. Traditional ML models are highly susceptible: with both added features, Random Forest and XGBoost become nearly perfect (F1 ≈ 1.0), which is implausible for a realistic fraud task. GNNs also improve, but to a smaller degree, and exhibit condition-specific sensitivity (notably GraphSAGE under +Failed7d). Overall, these results validate that the clean-baseline pipeline used for the main study better reflects true generalisation.

At the same time, these controlled ablations suggest an upper bound: if similarly informative and deployment-safe expert-engineered risk signals are available at inference time, they could materially improve detection performance. The pattern of results also indicates that the traditional ML baselines were not primarily improving through broader generalisation; instead, they appear to be exploiting the strong predictive relationship embedded in these added features, which drives their scores to near-perfect levels. By contrast, the GNN models improve more moderately under the same conditions, suggesting that their gains are less dependent on these shortcut signals and more consistent with learning from relational structure in the graph.

8 Conclusion and Future Work

This study demonstrates that heterogeneous GNN-based fraud detection outperforms traditional ML classifiers in clean-baseline recall and F1-score when trained on the same feature set with equivalent imbalance-handling strategies. On the held-out test set, HGT achieves the strongest GNN result (F1 = 0.416, recall = 0.576), improving over the best traditional baseline (Logistic Regression) by about 9% in F1 and 18% in recall. Among traditional models, tree-based ensembles collapse to near-zero recall under class-weight balancing, while linear models (LR, SVM) remain the most competitive non-graph baselines. The key driver of GNN performance is the ability to propagate relational context, particularly user-level fraud history, through message passing.

Future work should address the following: (i) richer graph schemas connecting transactions that share IP addresses or devices; (ii) temporal or dynamic GNN models that capture evolving behavioural patterns; and (iii) evaluation on real-world benchmarks such as IEEE-CIS Fraud Detection [10] to assess generalisation.

Table 4: Ablation study: F1-score under each feature condition.

Model	Clean	+Risk	+Failed7d	+Both
Random Forest	0.003	0.646	0.764	1.000
Extra Trees	0.025	0.634	0.764	0.982
XGBoost	0.354	0.622	0.744	0.999
Logistic Regression	0.381	0.549	0.629	0.706
SVM	0.381	0.549	0.629	0.707
GraphSAGE	0.369	0.491	0.079	0.491
GAT	0.390	0.402	0.465	0.451
HGT	0.416	0.438	0.540	0.627

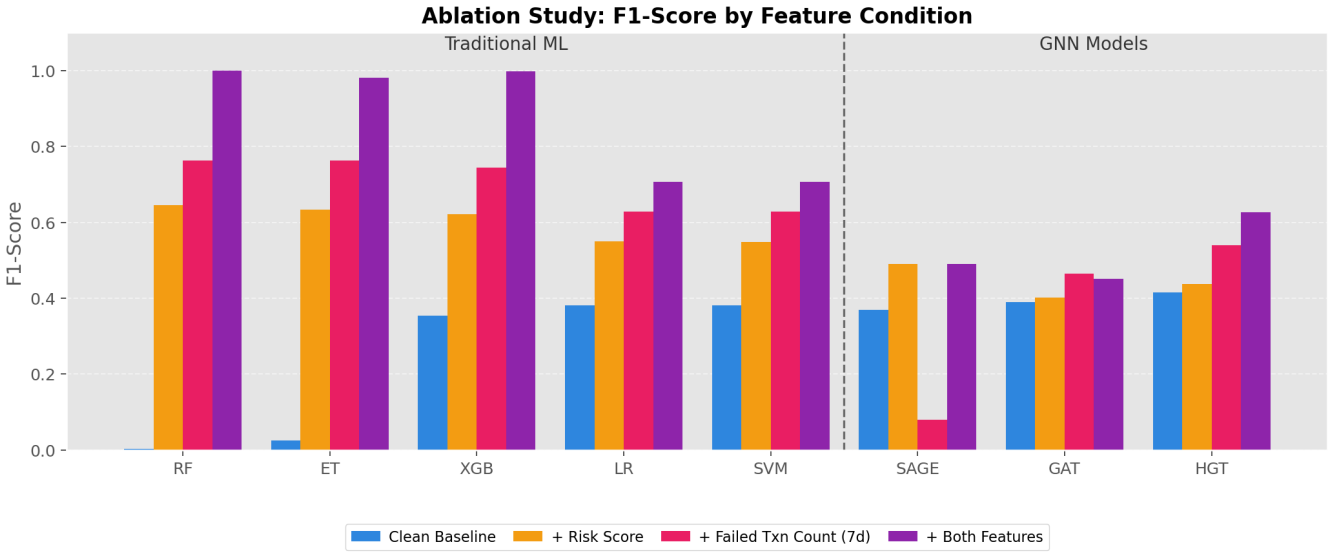

Figure 4: Ablation study: F1-score under four feature conditions. The dashed divider separates traditional ML from GNN models. Including both auxiliary risk features drives multiple models to near-perfect scores, indicating strong dependence on engineered risk signals.

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